



Hung Anh Trinh

FULLSTACK ENGINEER

Nguyen Trai, Thanh Xuan, Ha Noi

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Summary

- 8 years of solid backend development, and 2+ years of frontend experience.
- Regularly use Claude, ChatGPT, and GitHub Copilot for coding, ideation, and problem-solving.
- Strong Python expertise with practical experience in FastAPI, Django, and cloud deployment (GCP, Docker, Kubernetes).
- Experience building AI systems with vector databases, real-time inference, and scalable APIs.
- Collaborative problem-solver who translates requirements into technical solutions, identifies opportunities for improvement, and works well across teams.
- Focus on production quality: reliable deployments, monitoring, security, and system resilience across financial and high-volume services.

Technical Skills

Languages	Python (6+ years, expert), JavaScript (intermediate)
Python Internals	Metaclasses, import system, packaging internals
AI & LLM	Claude API, OpenAI SDK, LangChain, prompt engineering, vector embeddings
Backend Frameworks & APIs	FastAPI, Django, Flask, gRPC, Protocol Buffers
Developer Tooling & SDKs	SDK design, CLI tooling (Click, Typer), PyPI publishing, release pipelines
Type Safety & Testing	Strict typing (mypy, pyright), Pytest, SonarQube, integration testing
Databases	PostgreSQL, MongoDB, Redis, Milvus
Production & DevOps	Docker, Kubernetes, GCP, GitHub Actions, Celery, structured logging, CI/CD

Professional Experience

TIMS – SSI (Saigon Securities Inc.)

Hanoi, Vietnam

PYTHON BACKEND DEVELOPER

06/2023 – 03/2026

- Worked on TIMS (Treasury & Inventory Management System) for bond trading, credit limits, and cash management—contributed to business requirements translation and technical solutions.
- Implemented financial logic, data workflows, and forecasting algorithms; designed features that handled complex trading scenarios.
- Built real-time financial reporting dashboards with structured logging to support debugging and performance optimization.
- Maintained 85%+ test coverage with Pytest and SonarQube; helped team improve code quality practices.
- Designed authentication system with SSO integration and role-based access control (RBAC).
- Maintained background job system using Celery, Redis, and Kafka with reliability patterns for production use.

Genetica.asia

Hanoi, Vietnam

PYTHON BACKEND DEVELOPER

08/2022 – 06/2023

- Developed backend services and admin tools using Django REST framework; suggested improvements to API design based on integration feedback.
- Built user portal with React frontend integrated with backend APIs—worked closely with frontend team on requirements and API contracts.
- Implemented cart and payment system with third-party API integration, handling errors and edge cases.
- Refactored monolithic codebase into modular services and optimized PostgreSQL queries, improving performance by 40%+.
- Maintained API versioning, Swagger documentation, and test coverage with Pytest for code maintainability.

Delivery Service – Advessa Digital Solutions

Hanoi, Vietnam

PYTHON BACKEND DEVELOPER

09/2020 – 08/2022

- Built and maintained Django REST APIs for same-day delivery services—identified performance bottlenecks and implemented solutions, improving response times by 35%.
- Optimized system performance using Celery for async jobs, Redis for caching, and MongoDB for data storage—implemented reliability patterns like retry logic and rate-limiting.
- Implemented payment processing and commission logic; integrated third-party APIs with proper error handling and monitoring.
- Worked with frontend and mobile teams to deliver APIs with clear documentation and defined contracts.
- Deployed and maintained containerized services on Google Kubernetes Engine (GCP); set up CI/CD pipelines with GitHub Actions and structured logging for monitoring.

SMS System - IRIS Media Company

Ha Noi

DEVELOPER

10/2018 - 08/2020

- Built a Python GSM client with PySimpleGUI to manage device interactions, process backend commands, and deliver SMS with robust reporting.
- Crafted a Django backend exposing RESTful and callback APIs to orchestrate SMS workflows and monitor delivery.
- Optimized throughput and stability to support sending over 50,000 SMS per day while minimizing latency.
- Added proxy and channel fallback functions to reroute SMS traffic when on-premise GSM devices failed, preserving delivery continuity.
- Guided customers through API integration and troubleshooting to ensure smooth SMS delivery adoption.

Individual/Freelance Projects

Face Recognition System

Hanoi, Vietnam

AI ENGINEER

2024

- Built a face recognition system with vector embeddings and semantic search for real-time matching in events (1000+ faces).
- Indexed face embeddings in Milvus vector database; achieved 95%+ accuracy with sub-100ms query latency through optimization of indexing strategy.
- Built Gradio interface with FastAPI backend; deployed on on-premise servers with GPU acceleration (2080Ti).
- Used AI tools for rapid prototyping and implementation.

Captcha Recognition System

Hanoi, Vietnam

ML ENGINEER

2022

- Built captcha recognition using CNNs, RNNs, and LSTM; achieved 95%+ accuracy on easy/medium difficulty and 85%+ on hard captchas.
- Built FastAPI REST API for inference with request batching and async processing; implemented model versioning for reliability.
- Deployed and maintained system on on-premise servers with GPU acceleration (2080Ti); optimized inference latency and memory usage.

Education

HUST(Hanoi University of Science & Technology)

Bach Khoa, Hai Ba Trung, Ha Noi

SOFTWARE ENGINEERING

Aug. 2013 - June. 2018